Status	Finished
Started	Tuesday, 27 October 2026, 6:05 PM
Completed	Tuesday, 27 October 2026, 6:15 PM
Duration	9 mins 40 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

- a. The higher middle element is chosen as the middle element
- b. The lower middle element is chosen as the middle element ✕
- c. The middle element is chosen randomly
- d. The search stops

Question **2**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

- a. In the right sublist ✓
- b. At the beginning of the list
- c. In the left sublist
- d. In the middle of the list

Question **3**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

- a. It always starts from the beginning of the list
- b. It works on unsorted lists
- c. It compares elements sequentially
- d. It can be applied only if the list is sorted ✓

Question **4**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

- a. Compare the target element with the middle element in the list ✓
- b. Sort the list
- c. Compare the target element with the first element in the list
- d. Divide the list into two equal parts

Question **5**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

- a. Insertion
- b. Bubble ✓
- c. Selection
- d. Complexity

Question **6**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

- a. The list is sorted
- b. The search continues from the beginning
- c. The search stops
- d. The search continues in the left or right sublist ✓

Question **7**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

- a. n ✓
- b. log n
- c. n/2
- d. 1

Question **8**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(n log n)
b. O(log n)
c. O👎 ✓
d. O(1)

Question **9**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Interpolation search
b. Binary search
c. Linear search ✓
d. Hash search

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

a. Hashing
b. Dynamic search
c. Linear search ✓
d. Binary search

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search can be applied to both sorted and unsorted lists. ✓
b. Linear search is more efficient than binary search on large lists.
c. Linear search divides the list into halves.
d. Linear search requires the list to be sorted.

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It can only be applied to large lists
b. It is slower than linear search for small lists
c. It requires the list to be sorted ✓
d. It does not work with negative numbers

Question **13**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By dividing the list into halves
b. By using a hash function
c. By sorting the list first
d. By comparing each element in the list sequentially ✓

Question **14**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. An error is raised
b. The search is considered unsuccessful ✓
c. The last element is returned
d. The first element is returned

Question **15**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(log n)
b. O(n log n)
c. O
d. O(1) ✓